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SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

TOPIC 7: Practices for Text-Based Requirement Documentation(Part II)

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- Requirements Documents and Documentation Structures
- Prototypes in Requirements Engineering
- Quality Criteria for Work Products and Requirements

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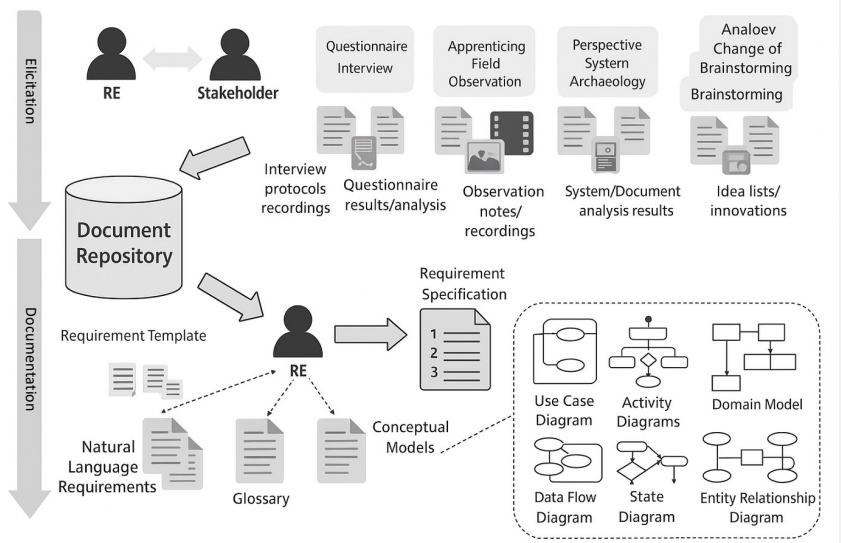
01 REQUIREMENTS DOCUMENTS AND DOCUMENTATION STRUCTURES

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Requirements Documentation

REQUIREMENTS DOCUMENT & STRUCTURE



innovative ● entrepreneurial ● global

Credit to: Mathias Lampe (2018)

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Requirements Documentation Structure

REQUIREMENTS DOCUMENT & STRUCTURE

- It is not sufficient to work with requirements at the level of individual requirements.
- Requirements must be collated and grouped in suitable work products
  - explicit requirements documents (eg. requirements specification)
  - other RE-related documentation structures (eg. product backlog)
- Document templates may be used to organize such documents with a well-defined structure (Refer to our previous lecture on Document templates)
- Well-defined structure helps to create a consistent and maintainable collection of requirements.

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Requirements Document

REQUIREMENTS DOCUMENT & STRUCTURE

Definition

A document in requirements engieering serves a specific purpose. Depending on their purpose, documents differ in terms of contents, format and quality

3 key characteristics of a document

Content	Format	Quality
• **Level of detail** • **Level of abstraction** • **Structure of document**	• **Textual** • **Model-Based** • **Hybrid**	• **Quality properties**

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Target Readers of Requirements Documents

REQUIREMENTS DOCUMENT & STRUCTURE



Clients, Contractors

Essential for:
- Contract, change management
- Planning of own activities
- Acceptance



Management

Essential for:
- Contract, change management
- Effort estimation, project planning, progress contril
- Resource planning
- Risk analysis



Development Team

Essential for
- Definition of the system architecture
- Implementation
- Preparation of the end user support materials



Tester

Essential for
- Specification of test cases

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Reasons for Documentation

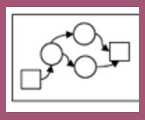
REQUIREMENTS DOCUMENT & STRUCTURE

1	**Basis of system development**	Influence other activities in project i.e analysis, design, implementation and testing
2	**Requirements documents are complex**	Handle complexity for anyone involved with suitable documentation
3	**Have a legal relevance**	Legally binding for contractor and client
4	**Requirements should be accessible to all**	Helps to understanding and supports new team members

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Requirements Representation Format

REQUIREMENTS DOCUMENT & STRUCTURE



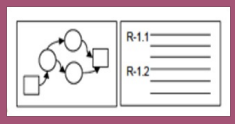
Model-Based

- Data perspective
- Functional perspective
- Behaviour perspective



Textual-Based

- Natural language text template



Hybrid (Combined)

- Model with anotations
- Text with models

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Frequently Used Requirements Documents

REQUIREMENTS DOCUMENT & STRUCTURE

01 **Vision document**

- A conceptual imagination of a future system,
- Describing its key characteristics and how it will create value for its users.

02 **Business requirements specification**

- The business goals, objectives, and needs of an organization that shall be achieved by employing a system (or a collection of systems)

03 **Stakeholder requirements specification**

- The stakeholders' desires and needs that shall be satisfied by building a system.
- Seen from the stakeholders' perspectives

04 **User requirements specification**

- Subset of a stakeholder requuirements specification
- Covering only requirements of stakeholders who are prospective users of a system.

05 **System requirements specification**

- The requirements for a system to be built and its context so that it satisfies its stakeholders' desires and needs

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Frequently Used Alternative Documentation Structures

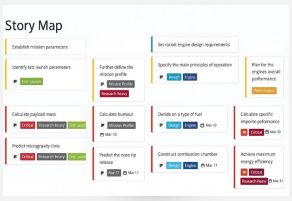
REQUIREMENTS DOCUMENT & STRUCTURE

- Used in agile projects e.g. in Scrum projects
- Agile projects often use a Vision document too



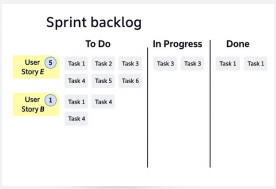
Product backlog

- A prioritized list of work items, covering all requirements that are needed and known for the product



Story Map

- A visual two-dimensional organization of user stories in a product backlog with respect to time and content



Sprint backlog

- A selected subset of a product backlog with work items that will be realized in the next iteration

Criteria for Selecting a Documentation Structure

REQUIREMENTS DOCUMENT & STRUCTURE

Both the selection of a documentation structure and internal organization of the chosen structure depend on

- **The chosen development process**
- **The development type and domain**
  - For example: Tailor made solution, product development, or standard product customizing
- **The contract**
  - A customer may prescribe the use of a given documentation structure
- **The size of the document**
  - The larger the document, the more structured is needed

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02 PROTOTYPES IN REQUIREMENTS ENGINEERING

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Prototype

PROTOTYPES IN REQUIREMENTS ENGINEERING

Definition

A preliminary, partial realization of certain characteristics of a system

Purposes of Prototype

Exploratory prototype

- Used to create shared understanding, clarify requirements or validate requirements
- Different level of fidelity
- Temporary work products that are discarded after use
- Also be used as a means of specification by example
  - must then be treated as evolving or durable work products.

Evolutionary prototype

- Pilot system that form the core of a system to be developed.
- The final system evolves by incrementally extending and improving the pilot system in several iterations,
- Agile system development frequently employs an evolutionary prototyping approach.

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Prototypes in Requirements Engineering

PROTOTYPES IN REQUIREMENTS ENGINEERING

Requirements Elicitation
- When stakeholders cannot express what they want clearly
- Helps to demonstrate what they would get, which help them shape their reuirements.

Use of exploratory prototypes as a means for:

Consider trade-off between cost and value gained
- Cost to build and use prototype
- Value gained in terms of easier elicitation and reduced risk of inadequate or even wrong requirements.
- Trade-off helps to decide which type of prptotype to use.

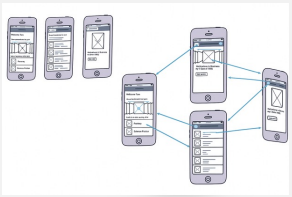
Validation
- Prototypes are a powerful means for validation the adequacy of requirements

Explore and decide about requirements variants for some aspect
- Eg. two different possible ways of supporting a given business process

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Types of Exploratory Prototypes

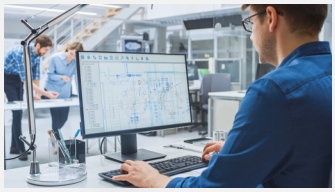
PROTOTYPES IN REQUIREMENTS ENGINEERING



Wireframes / Paper Prototypes
- Low-fidelity prototypes built with simple materials or sketching tools.
- Primarily for discussing and validating design ideas and user interface concepts



Mock-ups
- Medium-fidelity prototypes
- They use real screens and click flows, but without real functionality.
- Primarily for specifying and validating user interfaces



Native prototypes
- High-fidelity prototypes
- Implement critical parts of a system
- Stakeholders can use the prototype to see whether the prototyped part of the system will work and behave as expected.

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Tools to Create UI Prototypes

PROTOTYPES IN REQUIREMENTS ENGINEERING

- Balsamiq Wireframes
  https://balsamiq.com/wireframes/
- Marvel (also as app on phone)
  https://marvelapp.com/
- Framer
  https://www.framer.com/
- Adobe XD
  https://www.adobe.com/products/xd.html
- Sketch
  https://www.sketch.com/
- InVision
  https://www.invisionapp.com/

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03

QUALITY CRITERIA FOR WORK PRODUCTS AND REQUIREMENTS

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Importance of Quality Criteria

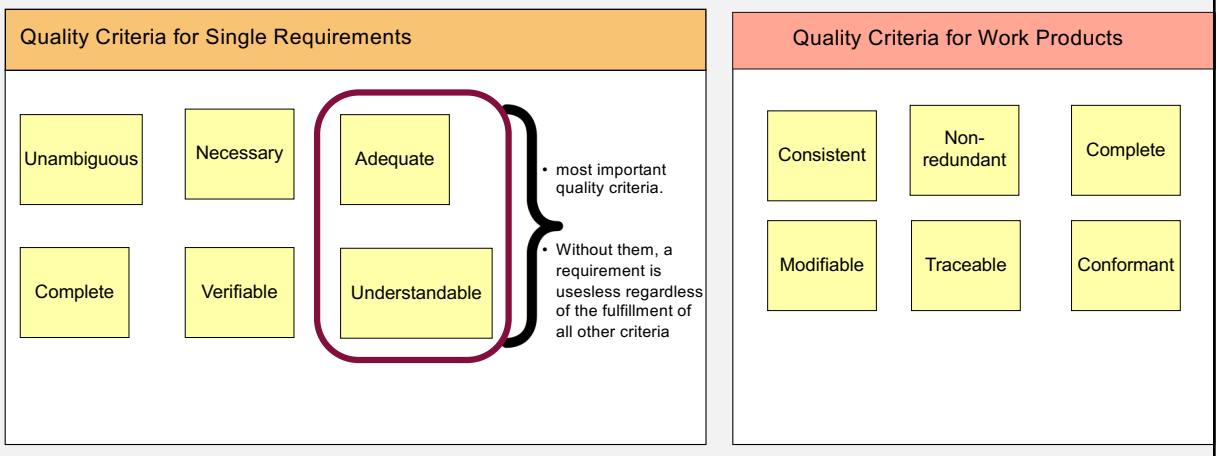
Quality Criteria for Work Products and Requirements

- Requirements need to meet specific quality criteria to be considered good requirements.
- Requirements do not have to adhere to all quality criteria completely.
  - Modern RE is value-oriented approach.
  - Degree of fulfillment for a quality criterion shall correspond to the value created by this requirement.
  - The higher the value of a requirement, the more relevant are the quality criteria, to reduce the risk of failure.
- Quality criteria should be defined
  - for single requirements AND
  - for sets of requirements (work products / document)

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Quality Criteria

Quality Criteria for Work Products and Requirements



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Quality Criteria for Single Requirement: Adequate

Quality Criteria for Work Products and Requirements

Definition

A single requirement is of adequate quality if it provides just enough detail to be understood and implemented correctly without being overly complex or ambiguous.
It must be **clear, specific, and testable**, but not overloaded with unnecessary technical details.

Example Adequate Requirement

Requirement:
"The system shall allow users to reset their password via a secure email verification link."

✅ Why it is adequate:
- Clear and understandable (reset password, via email verification).
- Provides enough detail to implement (secure link is required).
- Does not go into technical implementation specifics

Example Non-Adequate Requirement

Requirement:
"The system shall have good password reset functionality."

❌ Why it is not adequate:
- Vague ("good" is subjective and ambiguous).
- Lacks detail on how the reset will happen (email, SMS, security questions?).
- Cannot be tested or implemented reliably since the criteria for success are unclear.

Quality Criteria for Single Requirement: Understandable

Quality Criteria for Work Products and Requirements

Definition

A single requirement is of understandable quality if it is written in clear, simple, and unambiguous language so that all stakeholders interpret it in the same way.

Example Understandable Requirement

Requirement:
"The mobile app shall display the user's account balance on the dashboard after login."

✅ Why it is understandable:
- Clear, simple, and specific
- Interpreted the same by all stakeholders
- Testable (expected output and location are defined)

Example Non-Understandable Requirement

Requirement:
"The system shall provide relevant financial information in an appropriate place."

❌ Why it is not understandable:
- Vague terms: "relevant," "appropriate"
- Open to multiple interpretations
- Not testable (no measurable criteria)

Quality Criteria for Single Requirement: Unambiguous

Quality Criteria for Work Products and Requirements

Definition

A single requirement is of unambiguous quality if it is stated in precise terms that allow only one clear interpretation by all stakeholders

Example Unambiguous Requirement

Requirement:

"The system shall display search results within 2 seconds after the user presses the search button."

✅ Why it is understandable:

- Uses precise wording ("2 seconds" instead of "quickly").
- Only one possible interpretation for what is expected.
- All stakeholders (users, developers, testers) will agree on the same meaning.

Example Ambiguous Requirement

Requirement:

"The system shall display search results as soon as possible after the user presses the search button."

❌ Why it is ambiguous:

- "As soon as possible" can mean different things to different stakeholders (developer = 5s, user = <1s).
- Leaves room for multiple interpretations.
- Cannot be tested reliably because no clear performance measure is defined.

Quality Criteria for Single Requirement: Complete

Quality Criteria for Work Products and Requirements

Definition

A single requirement is of complete quality if it provides all the necessary information, conditions, and constraints so that it can be fully implemented and tested without needing further clarification.

Example Complete Requirement

Requirement:

"The system shall send a password reset link to the user's registered email address within 1 minute after the user submits a valid 'Forgot Password' request, and the link shall remain valid for 24 hours."

✅ Why it is complete:

- States the trigger (user submits a valid request).
- Defines the action (send reset link to registered email).
- Specifies timing (within 1 minute).
- Includes constraint (link valid for 24 hours).
- Can be fully implemented and tested without extra clarification.

Example Incomplete Requirement

Requirement:

"The system shall allow users to reset their password."

❌ Why it is incomplete:

- "Missing trigger (how does the reset start?).
- Missing delivery method (email, SMS, security questions?).
- No timing defined (immediately, within seconds, hours?).
- No constraints on validity or security.
- Leaves gaps that developers and testers would need to clarify.

Quality Criteria for Single Requirement: Necessary

Quality Criteria for Work Products and Requirements

Definition

A single requirement is of necessary quality if it specifies a feature or condition that directly contributes to fulfilling stakeholder needs or system objectives, without adding superfluous or redundant functionality.

Example Necessary Requirement

Requirement:

"The system shall require users to authenticate with a valid username and password before accessing their account."

✅ Why it is necessary:

- Directly supports a core stakeholder need: security.
- Essential for protecting user data and meeting system objectives.
- Not redundant, since without it the system would fail to ensure secure access.

Example Unnecessary Requirement

Requirement:

"The system shall play background music on the login page."

❌ Why it is unnecessary

- Does not contribute to the core system objectives (security, usability, functionality).
- Adds unnecessary complexity and potential distractions for users.
- Increases development and maintenance cost without clear stakeholder need.

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Quality Criteria for Single Requirement: Verifiable

Quality Criteria for Work Products and Requirements

Definition

A single requirement is of verifiable quality if it can be objectively tested, measured, or checked through inspection, or testing to confirm that it has been correctly implemented.

Example Verifiable Requirement

Requirement:

"The system shall lock a user's account after five consecutive failed login attempts."

✅ Why it is verifiable

- Can be tested directly by attempting five failed logins.
- Has clear, measurable criteria ("five consecutive attempts").
- Provides an objective pass/fail outcome for testers.

Example Non Verifiable Requirement

Requirement:

"The system shall provide a user-friendly login process."

❌ Why it is non-verifiable

- "User-friendly" is subjective and cannot be measured objectively.
- Different users may have different opinions on what is "friendly."
- No clear test criteria to determine pass or fail.

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Quality Criteria for Requirement Specification: Complete

Quality Criteria for Work Products and Requirements

- The set of requirements must be complete.
- If requirements are missing, then the product will also be incomplete.
- One way to find missing requirements is to have a checklist of the kinds of requirements you have created on similar projects, or to which similar systems are subject to.
- Having domain experts review your requirements with you will also helpful.
- When one requirement is changed, similar requirements are most likely close at hand.

Quality Criteria for Requirement Specification: Consistent

Quality Criteria for Work Products and Requirements

- The set of requirements must be consistent.
- If there are requirements in your specification that conflict with each other, then the product will not be able to meet all your requirement.
- Inconsistencies must be resolved for your project to succeed.

Quality Criteria for Requirement Specification: Consistent

Quality Criteria for Work Products and Requirements

Here is one requirement:

"The system shall display all reports on the user screen within 10 seconds of the user pressing the "enter" key."

Here is a related requirement for the same system:

"The system shall display the Aging Report on the user screen within 30 seconds of the user pressing the "enter" key."

What is the problem with the 2 set of above requirements? Is it consistent?

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Quality Criteria for Requirement Specification: Modifiable

Quality Criteria for Work Products and Requirements

If you must change a requirement (create, edit or delete), you must be able to evaluate the impact of that change on all other requirements.

Organize your requirements into categories by their scope or purpose. When one requirement is changed, similar requirements are most likely close at hand.

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Quality Criteria for Requirement Specification: Traceable

Quality Criteria for Work Products and Requirements



You must be able to tie a derived requirement back to its origin (source, specific stakeholder that request it during elicitation).



You must be able to trace a requirement to the hardware/software that implements it, as well as to the test cases that verify it.

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Quality Criteria for Requirement Specification: Non-Redundant

Quality Criteria for Work Products and Requirements

- Each requirement should appear only once to prevent inconsistencies and conflicting interpretations.
- Non-redundant requirements ensure developers and testers know exactly where to refer, reducing confusion
- A non-redundant document is easier to update, since changes only need to be made in one place.

Example Non-Redundant Requirement

Requirement:

"The system shall send a confirmation email after a successful user registration."

✅ Why it is complete:
- Only one clear statement of behavior
- Easy to maintain and test

Example Redundant Requirement

Requirement:

"The system shall notify the user by email after registration is successful."

"The system shall send a confirmation email when a user registers successfully."

❌ Why it is redundant:
- Same requirement repeated in different wordings
- Creates risk of inconsistency during updates

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Quality Criteria for Requirement Specification: Conformant

Quality Criteria for Work Products and Requirements

- Each requirement must follow defined templates, style guides, or organizational standards (e.g., IEEE SRS format)
- Conformant requirements use uniform structure, terminology, and notation so the document is easy to read and review
- A conformant format ensures requirements can be traced, validated, and compared systematically during reviews and audits

Example Conformant Requirement

Requirement:
"The system shall log out a user automatically after 15 minutes of inactivity." (written in IEEE-style: "The system shall …" + measurable condition)

✅ Why it is conformant:
- Follows standard structure ("The system shall…")
- Uses precise, measurable value (15 minutes)
- Easy to validate and trace in SRS

Example Non-Conformant Requirement

Requirement:
"Users will get logged out when they are idle for some time."

❌ Why it is non-conformant:
- Informal, inconsistent phrasing
- Vague term "some time"
- Hard to test and inconsistent with standards

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THANK YOU

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